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A2

1)

a) $\lim_{x \to 0} \frac{x}{\sqrt{x^2+x^4}}$

1º $\sqrt{x^2+x^4}$

$\downarrow$

x^2+x^4 simplificando: $x^2(1+x^2)$

2º $\sqrt{x^2(1+x^2)}$

$\downarrow$

$\sqrt{x^2}\sqrt{x^2+1}$

como $\sqrt{x^2} = x$

Assim:

$x\sqrt{x^2+1}$

Agora substituindo:

$\lim_{x \to 0} \frac{\cancel{x}}{\cancel{x}\sqrt{x^2+1}} =$

$= \lim_{x \to 0} \frac{1}{\sqrt{x^2+1}} =$

$= \lim_{x \to 0} \frac{1}{\sqrt{0^2+1}} = 1$

b) $\lim_{x \to 0} x^5 \cdot \cos \frac{1}{x} = \lim_{x \to 0} x^4 \cdot \cancel{x} \cdot \cos \frac{1}{\cancel{x}} = \lim_{x \to 0} x^4 \cdot \cos$

$= \lim_{x \to 0} 0^4 \cdot \cos = \lim_{x \to 0} 0 \cdot \cos = 0$

c) $\lim_{x \to \infty} \left(1 + \frac{1}{20x}\right)^x$

$\lim_{x \to \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$

$\lim_{x \to \infty} \left[\left(1 + \frac{1}{20x}\right)^{20x}\right]^{\frac{1}{20}} = e^{\frac{1}{20}}$

d) $\lim_{x \to 0^-} \frac{|x|}{x}$ → módulo

1º limites laterais

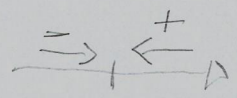
negativa quando $x \to 0$

$\lim_{x \to 0^-} -\left(\frac{-x}{x}\right)$

$\lim_{x \to 0^-} = -(-1) = -1$

e) $\lim\limits_{x \to 2} \dfrac{|x-2|}{x-2} = \lim\limits_{x \to 2} \dfrac{|2-2|}{2-2} = \dfrac{|0|}{0}$ indet

$$\left.\begin{array}{l} \lim\limits_{x \to 2^+} \left(\dfrac{|x-2|}{x-2}\right) = 1 \\ \\ \lim\limits_{x \to 2^-} \left(\dfrac{|x-2|}{x-2}\right) = -1 \end{array}\right\} \text{ diferentes}$$



$f(x) = \dfrac{|x-2|}{x-2}$

$\lim\limits_{x \to 2^-} f(x) \neq \lim\limits_{x \to 2^+} f(x)$

ou seja

$$\boxed{\lim_{x \to 2} \frac{|x-2|}{x-2} = \nexists}$$

f) $\lim\limits_{x \to 1} \left(\dfrac{x^2+x-2}{x^2-x}\right) = \lim\limits_{x \to 1} \dfrac{1^2+1-2}{1-1} = \dfrac{0}{0}$ indet

fatorar

$x^2 + x - 2 = (x-1).(x+2)$

$x^2 - x = x(x-1)$

Assim:

$$\lim_{x \to 1} \frac{\cancel{(x-1)}.(x+2)}{x\cancel{(x-1)}} = \lim_{x \to 1} x+2 = \boxed{\lim_{x \to 1} 3 = 3}$$

g) $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \lim_{x \to 0} \frac{1 - \cos 0}{0^2} = \frac{1-1}{0} = \frac{0}{0}$ indet

usando o princípio fundamental trigonométrico

$\lim_{x \to 0} \frac{\operatorname{sen} x}{x} = 1$

sabendo que

$\operatorname{sen}^2 x + \cos^2 x = 1$

e

$\operatorname{sen}^2 x = 1 - \cos^2 x$

multiplicar pela conjugada

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} \cdot \frac{(1 + \cos x)}{(1 + \cos x)} = \lim_{x \to 0} \frac{1 - \cos^2 x}{x^2 (1 + \cos x)}$$

Substituindo $\operatorname{sen}^2 x = 1 - \cos^2 x$:

$$\lim_{x \to 0} \frac{\operatorname{sen}^2 x}{x^2 (1 + \cos x)}$$

simplificando :

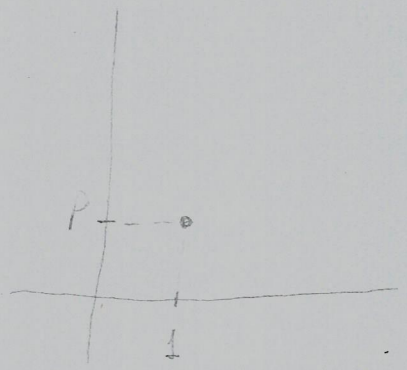
$$\lim_{x \to 0} \underbrace{\frac{\operatorname{sen} x}{x}}_{\text{fundamental}} \cdot \underbrace{\frac{\operatorname{sen} x}{x}}_{\text{fundamental}} \cdot \frac{1}{1 + \cos x} = \lim_{x \to 0} \frac{1}{1 + \cos 0}$$

$$\lim_{x \to 0} \frac{1}{1+1} = \frac{1}{2}$$

2)

$$f(x) = \begin{cases} \dfrac{x-1}{x^2-1} & \text{se } x \neq 1 \\ p & \text{se } x = 1 \end{cases}$$

$f(1) = P$ Existe



$x \neq 1$

$$\lim_{x \to 1} \frac{x-1}{x^2-1} = \frac{\cancel{x-1}}{(x+1)\cdot\cancel{(x-1)}} = \lim_{x \to 1} \frac{1}{x+1} = \frac{1}{2} \text{ Existe}$$

$x = 1$

$f(1) = p$

$\lim_{x \to 1} f(x) = \frac{1}{2}$

$\neq 1$

É possível que a função na parte "p" seja contínua, para isso a $f(x) = \lim_{x \to a} f(x)$ devem ser iguais, ou seja $f(1)$ deve ser $\frac{1}{2}$.

Condições para que "exista" continuidade em determinado ponto:

(a) $f(a)$ existe

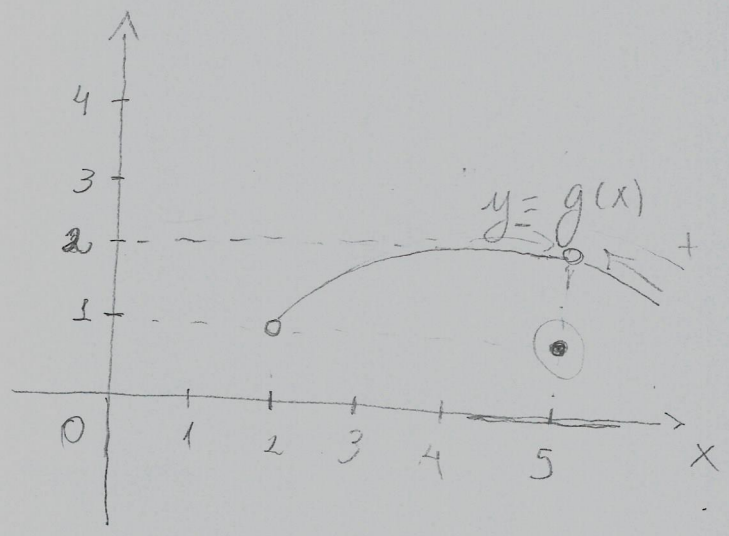
(b) $\lim f(x)$ existe

(c) $\lim_{x \to a} f(x) = f(x)$

3) $g(x)$

$x = 5$

$D =]2, 5]$



(a) $g(5) = 1$

limites laterais

(b) $\lim_{x \to 5^+} g(x) = \lim_{x \to 5^+} = 2$

$\lim_{x \to 5^-} g(x) = \lim_{x \to 5^-} = 2$

Existem

(c) $g(5) \neq \lim_{x \to 5} g(x)$

como são diferentes, não existe continuidade no $x = 5$.